Alexandria University
Faculty of Engineering

Natural Language Processing
Assigned: 26/10/2025
Due: 05/11/2025

# Assignment 2

## Problem Definition

In this assignment, you will implement the **Word2Vec** model using **SkipGram with Negative Sampling** to build learned embeddings. You will also use them for building a **Neural Network** and **Hidden Markov Model** for **Named Entity Recognition Task.**

## Part 1

You are required to implement a **Word2Vec model** using the **Skip-Gram with Negative Sampling (SGNS)** approach. Follow the steps below carefully:

1. **Dataset**
    - Use the **lhoestq/conll2003** dataset from **Hugging Face**.
    - Load the dataset and extract all text data (train, validation, and test sets).
    - Use the **entire dataset** for training your model.
2. **Preprocessing**
    - Apply the needed preprocessing.
3. **Model Implementation**
    - Implement the **Skip-Gram** architecture where the model predicts context words given a target word.
    - Use **Negative Sampling** to efficiently approximate the softmax output and speed up training.
4. **Training**
    - Train your model on the full corpus until convergence.
    - Monitor loss to ensure stable learning.
5. **Saving Embeddings**
    - After training, **save the learned word embeddings**.

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- These embeddings will be used in the **next part of the assignment**.

6. **Word Analogy Function**
    - Implement a **function for word analogy**, which should take three input words (e.g., *A,B.A**) and predict the fourth word (*B**).

# Part 2

In this part of the assignment, you will use the **word embeddings learned in Part 1** to build and evaluate **Named Entity Recognition (NER)** models using **two different approaches**:

1. A **Neural Network–based model**
2. A **Hidden Markov Model (HMM)**–based model

## NER Using Neural Network

- Implement a **Feed-Forward Neural Network model**
- Use the **three splits** of the data:
    - **Train set** → for training the model
    - **Validation set** → for tuning hyperparameters
    - **Test set** → for final evaluation
- The input to the model should be the **word embeddings** from Part 1.
- The output should be the **predicted entity tags** for each token.
- Evaluate the model using metrics such as **accuracy**, **precision**, **recall**, and **F1-score**.

## NER Using Hidden Markov Model (HMM)

- Implement a **Hidden Markov Model**–based NER system.
- Use only the **train split** for training and the **test split** for evaluation.
- Compute transition and emission probabilities from the training data.
- Use the **Viterbi algorithm** for decoding the most likely sequence of entity tags for the test sentences.
- Evaluate the performance using metrics such as **accuracy**, **precision**, **recall**, and **F1-score**.
- Compare results to the **Feed-Forward Neural Network model**.



## Deliverables

- This project must be done in teams of three or four.
- Teams of two are **NOT** allowed.
- You should deliver a notebook containing all your code, plots, results and comparisons.

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